

AgentBook Retrieval-Augmented Learning System: Sample Technical Notes

A synthetic English document for testing RAG, summarization, study-guide generation, table retrieval, and cross-lingual Vietnamese-to-English queries.

Abstract

This document describes a compact retrieval-augmented learning system named AgentBook. It is intentionally built as a text-based PDF with headings, paragraphs, tables, and figures. The file is suitable for testing document parsing, chunking, embedding, vector search, answer generation, summarization, and study-guide endpoints.

The central idea is simple: users upload learning materials, the parser converts heterogeneous content into normalized blocks, and a retrieval layer selects evidence for downstream agents. Even though this document is synthetic, its structure imitates a technical paper so that automated test cases can exercise realistic layouts.

1. System Overview

AgentBook ingests materials from PDF, DOCX, PPTX, image, spreadsheet, and audio sources. Each material is assigned to a collection so that experiments can isolate ordinary text retrieval from OCR-heavy and table-heavy workflows. The system stores both raw block metadata and vector representations, enabling semantic search as well as debugging through material-level diagnostics.

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1.1 Core Components

Component	Primary role	Expected debug signal
Parser Router	Selects Docling, OCR, spreadsheet, or audio parser	parser_name and modality
Chunk Builder	Creates retrieval-ready blocks	chunk_count > 0
Vector Indexer	Embeds chunks and writes them to Qdrant	qdrant_vector_count > 0
Query Agent	Generates grounded answers from retrieved evidence	citations and source block IDs
Study Guide Agent	Converts source content into learning objectives and quiz items	guide sections and key concepts

Table 1. Core components and expected debug signals.

2. Retrieval Pipeline

The retrieval pipeline begins with material normalization. Text blocks, table blocks, figure captions, OCR blocks, and audio transcript segments are converted into a common schema. This schema preserves the source page, sheet, timestamp, block type, and confidence score when applicable. The normalized schema is essential because downstream agents should not care whether a fact came from a PDF paragraph or a spreadsheet row.

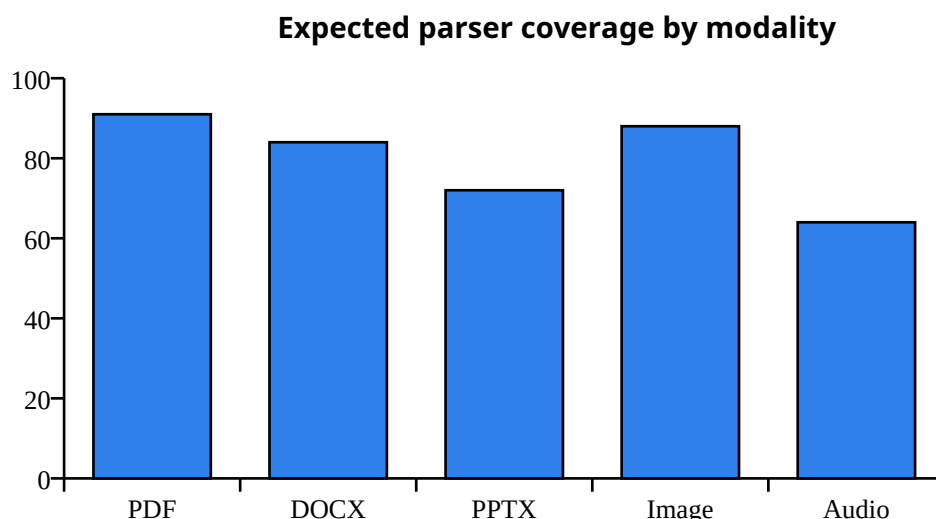


Figure 1. Synthetic coverage score used for parser smoke testing.

For cross-lingual testing, a user may ask a Vietnamese question over this English document. A robust system should retrieve the relevant English evidence, then answer naturally in Vietnamese while preserving factual grounding. The test is not about translation alone; it is about whether retrieval remains stable when the query language differs from the source language.

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3. Summarization and Study Guide Tasks

The summarize endpoint should produce a concise overview of the material, including system purpose, input modalities, parser behavior, and debug signals. The study-guide endpoint should be more pedagogical: it should list learning objectives, define key terms, propose review questions, and highlight practical implementation risks such as low-quality OCR images or incorrectly chunked tables.

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4. Evaluation Plan

Test case	Input material	Success criterion
Vietnamese ask over English PDF	lecture_notes.pdf	Answer in Vietnamese with evidence from English text
Table retrieval	DOCX or CSV table	Returns exact row or aggregated value
OCR pass	clear scan image	OCR confidence available and content searchable
OCR fail	blurred handwriting image	System refuses or reports quality error
Compare	PDF + DOCX same topic	Identifies overlapping and different claims

Table 2. Evaluation plan for the AgentBook sample pack.

The most important evaluation rule is that a parser should fail loudly when the input is too poor to process. Silent failure is worse than refusal because it allows hallucinated answers to appear trustworthy. For this reason, the test pack includes both pass and fail examples.

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5. Practical Notes and Expected Questions

Question: What is the role of the vector indexer?

A correct answer should cite the relevant block, avoid unsupported claims, and preserve the answer language requested by the user.

Question: Which endpoint can create a study guide?

A correct answer should cite the relevant block, avoid unsupported claims, and preserve the answer language requested by the user.

Question: Why should the system reject low-quality OCR images?

A correct answer should cite the relevant block, avoid unsupported claims, and preserve the answer language requested by the user.

Question: How does the system support compare across two documents?

A correct answer should cite the relevant block, avoid unsupported claims, and preserve the answer language requested by the user.

Question: What debug signal confirms that vectors were written?

A correct answer should cite the relevant block, avoid unsupported claims, and preserve the answer language requested by the user.

End of sample text-based PDF.